EASTERN INTERNATIONAL UNIVERSITY
BECAMEX BUSINESS SCHOOL



ASSIGNMENT 1
PERSONAL PORTFOLIO SITE

Course: MIS 311 - Introduction to Business Analytics
Lecturer: Ms. Nguyễn Thị Quỳnh Như

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TABLE OF CONTENT

1. Data Overview..... 3
2. Data Cleaning..... 4
2.1 Missing Values..... 4
2.2 Duplicate Records..... 4
3. Descriptive analysis..... 5
3.1: Income vs. Cost of Living Baseline..... 5
3.2 Regional Comparison..... 7
4. Conclusion..... 7

1. Data Overview

Item	Description
Dataset Name	Cost of Living
Source	The dataset provided by the lecturer from MIS 311 - Introduction from Business Analytics
Number of rows	201
Number of columns	5
Variables	Country, Year, Average Monthly Income, Cost of Living, Region

Table 1: Dataset overview

Variable	Data Type	Description
Country	Text	The name of the country corresponding to the economic data record.
Year	Numerical	The specific year in which the data was recorded (ranging from 2000 to 2023).
Average Monthly Income	Float	The average monthly income level (measured in USD or standard currency units).
Cost of Living	Float	The index or monetary value representing the overall cost of living.
Region	Text	The global geographical region where the country is located (e.g., Asia, Europe, Africa).

Table 2: Variable description

The Cost of Living dataset is designed to provide information about average monthly income and living expenses from different countries and geographical regions. It has 201 records and 5 variables. The dataset consists of 12 countries from 2000 to 2023 that have differences in income and living costs. This helps the data analysts examine how these values vary across years. By analysing these variables, we can identify patterns, compare income levels with living expenses, and gain insights into the potential affordability of living in different countries. Before conducting the analysis, the dataset needs to be checked for

missing values and duplicate records to ensure that the results are as reliable as possible. The original data source should also be identified and acknowledged.

2. Data Cleaning

2.1 Missing Values

Variable	Missing values	Percentage	Action taken
Average_Monthly_Income	2	1.01%	No action needed
Region	2	1.01%	Fill with the correct region

Table 3: Missing Value Summary

The dataset contained four missing values: two in the Average_Monthly_Income column and two in the Region column. The missing Region values were filled with "North America" based on the countries' geographical locations. The missing income values were left blank because the actual values could not be confirmed, avoiding inaccurate data.

2.2 Duplicate Records

Item	Results
Duplicate Rows Found	2
Action Needed	Remove

Table 4: Duplicate Records Summary

Two duplicate records were identified and removed using Excel's Remove Duplicates function. This reduced the dataset from 201 to 199 records and helped prevent duplicate entries from affecting the analysis.

3. Descriptive analysis

3.1 Income vs. Cost of Living Baseline

Descriptive analytics summarizes raw data to provide a clear understanding of its central trends, overall distribution, and spread

Average_Monthly_Income		*Cost_of_Living*	
Mean	4243,967157	**Mean**	3705,127286
Standard Error	151,5715864	**Standard Error**	140,515613
Median	4266,46	**Median**	3688,09
Mode	#N/A	**Mode**	#N/A
Standard Deviation	2127,408594	**Standard Deviation**	1982,216654
Sample Variance	4525867,326	**Sample Variance**	3929182,864
Kurtosis	-1,185542056	**Kurtosis**	-1,272006829
Skewness	0,055355678	**Skewness**	-0,04103302
Range	7441,82	**Range**	6516,53
Minimum	534,74	**Minimum**	464,49
Maximum	7976,56	**Maximum**	6981,02
Sum	836061,53	**Sum**	737320,33
Count	197	**Count**	199
Confidence Level(95,0%)	298,9205736	**Confidence Level(95,0%)**	277,0992448

Table: 5: Descriptive Statistics for Monthly Income and Cost of Living

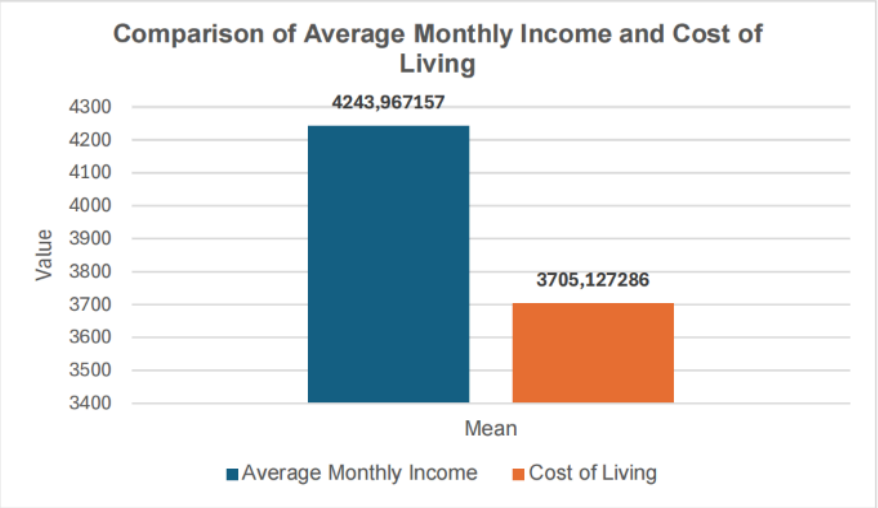


Figure 1: Comparison of Average Monthly Income and Cost of Living

From Table 5 and Figure 1, the average monthly income (4,243.97) is higher than the average cost of living (3,705.13). This means that, on average, people earn enough to cover their basic living expenses with some money left over. However, because both values are close, many individuals might still face financial pressure depending on where they live.

Additionally, economic fluctuations like inflation or unexpected expenses can quickly put individuals under financial strain, especially in high-cost regions.

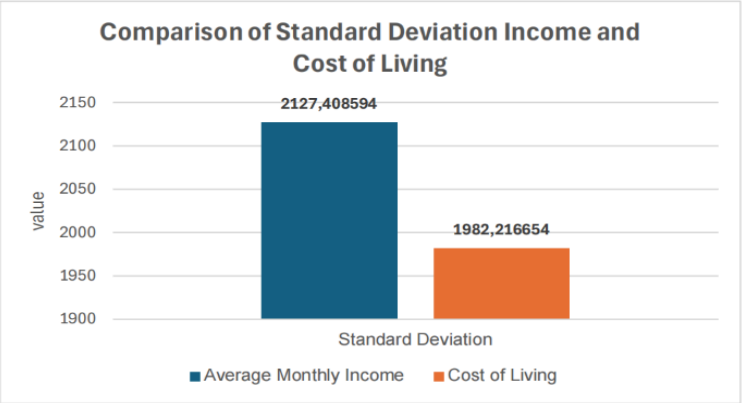


Figure 2: Comparison of Standard Deviation Monthly Income and Cost of Living

From Table 5 and Figure 2, Both average monthly income and cost of living exhibit very high standard deviations with 2,126.41 and 1,982.22 respectively. This shows that the data points are widely spread out rather than closely clustered around the average. This high variation highlights significant global economic inequality, meaning that national income levels and living costs vary drastically across different countries and regions.

3.2 Regional Comparison

Row Labels	Average of Average_Monthly_Income	Average of Cost_of_Living
Africa	4695,19	3963,17
Asia	4172,51	3645,29
Europe	4477,26	3554,35
North America	3764,28	3750,07
Oceania	4548,39	3913,40
South America	4389,57	3585,02
Grand Total	**4243,97**	**3705,13**

Table 6: Regional Comparison of Average Monthly Income and Cost of Living

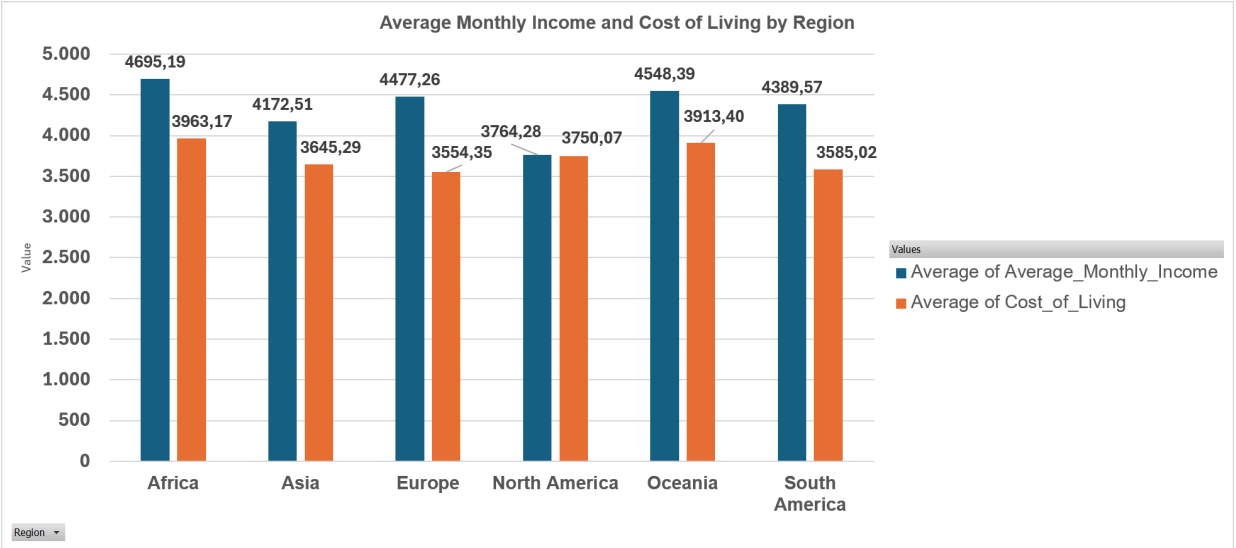


Figure 3: Regional Comparison of Average Monthly Income and Cost of Living

Looking across regions, North America has the narrowest difference between income and living expenses ($3,764.28$ vs. $3,750.07$, leaving only a 14.21 surplus), whereas Europe enjoys the largest financial gap with an average margin of over 922.91.

This demonstrates that geographic location significantly impacts net savings potential, meaning individuals in regions like North America face much tighter financial flexibility compared to those in Europe.

4. Conclusion

In conclusion, this data analysis highlights significant global economic disparities, showing that while average incomes slightly exceed living costs, high standard deviations reflect vast inequality across countries. Furthermore, regional comparisons indicate that geographic location heavily influences financial stability, with regions like North America experiencing much tighter disposable margins than Europe. Overall, these findings provide a clear understanding of global purchasing power dynamics and underscore the financial vulnerabilities populations face in high-cost environments.